HUNDRED DAYS OF CODE
DAY 6

Question 1:

Write a C program to find the remaining area of a triangular field when a square pond of given side is removed from it.

STEP 1: UNDERSTANDING AND IDENTIFYING THE REQUIRED INPUT, OUTPUT, AND FORMULA.

- We have a triangular field and a square pond of some dimension and we need to identify the remaining area of the triangular field if the square pond is removed from it.
- INPUT: we will be in need of:

  Base and height of triangular field.

  Side length of the square pond.

- OUTPUT: it will be the remaining required area of the triangular field.
- FORMULA:

  Area of triangle: $\mathrm{A} = \frac{1}{2} \times base \times height$

  Area of square pond: $\mathrm{S} = side \times side$

  Remaining area: R = A – S

STEP 2 :- **TEST CASE TABLE:**

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT (PASS/FAIL)
1	BASE = 20, HEIGHT = 8, SIDE= 4	64.00 sq. units	64.00 sq. units	pass
2	BASE= 6, HEIGHT=5, SIDE=2	11.00 sq. units	11.00 sq. units	pass
3	BASE=12, HEIGHT=7, SIDE=5	17.00 sq. units	17.00 sq. units	pass

STEP 3 : ALGORITHM

1. Start.
2. Declare variables for the triangle's base, height, and square side.
3. Read the base, height, and side of the square pond.
4. Calculate the area of the triangle using:
   triangle area = 0.5 × base × height
5. Calculate the area of the square using:
   pond area = side × side

6. Calculate the remaining area:
   remaining area = triangle area – pond area
7. Display the remaining area.
8. Stop.

The algorithm correctly:

- Accepts all required inputs.
- Calculates both areas using the correct formulas.
- Subtracts the pond area from the triangular field area.
- Produces the required remaining area.

STEP 4: REQUIRED C PROGRAM

```
#include <stdio.h>

int main(void) {
  double base, height, side;
    double triangle area, pond area, remaining area;

    printf("Enter triangle base, triangle height, and square side: ");
    scanf("%lf %lf %lf", &base, &height, &side);

    triangle_area = 0.5 * base * height;
    pond_area = side * side;
    remaining_area = triangle_area – pond_area;

    printf("Remaining area = %.2f square units\n", remaining_area);

    return 0;
}
```

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STEP 5: COMPILATION, EXECUTION, AND EVALUATION

```
┌──(kali㉿kali)-[~]
└─$ cc hdoc6.c

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter triangle base, triangle height, and square side: 20 8 4
Remaining area = 64.00 square units

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter triangle base, triangle height, and square side: 6 5 2
Remaining area = 11.00 square units

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter triangle base, triangle height, and square side: 12 7 5
Remaining area = 17.00 square units

┌──(kali㉿kali)-[~]
└─$ 
```

all the three actual outputs match their expected outputs.

Overall result: PASS.

Question 2:

Write a C program to find the perimeter, area, and each interior angle of a regular polygon when its number of sides, side length, and apothem are given.

STEP 1: UNDERSTANDING AND IDENTIFYING THE REQUIRED INPUT, OUTPUT, AND FORMULA.

- we are given number of sides , length of each side and apothem of a polygon and we need to find its perimeter , area, and each interior angle of the respective polygon .

- INPUT: number of sides = n
  Side length of polygon = s
  Apothem = a

- OUTPUT : perimeter, area, and each interior angle.

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-FORMULA REQUIRED: **Perimeter:** $P = n \times s$

Area: $A = \frac{1}{2} \times P \times a$

or

$$A = \frac{1}{2} \times n \times s \times a$$

Each interior angle: *I = n(n−2)×180* / n

STEP 2: TEST CASE TABLE:

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT	RESULT
1	n=5, side=4,apothem=2.75	P = 20.00, A = 27.50, Angle = 108.00°	P = 20.00, A = 27.50, Angle = 108.00°	pass
2	n=6, side=3, apothem=2.598	P = 18.00, A = 23.38, Angle = 120.00	P = 18.00, A = 23.38, Angle = 120.00°	pass
3	n=8, side=5, apothem=6.036	P = 40.00, A = 120.72, Angle = 135.00°	P = 40.00, A = 120.72, Angle = 135.00°	pass

STEP 3: ALGORITHM

1. Start.
2. Declare variables for number of sides, side length, and apothem.
3. Read the number of sides, side length, and apothem.
4. Calculate the perimeter: perimeter = n × side
5. Calculate the area: area = 0.5 × perimeter × apothem
6. Calculate each interior angle:
interior_angle = ((n - 2) × 180) / n
7. Display the perimeter, area, and interior angle.
8. Stop.

The algorithm uses the standard formulas for a regular polygon and calculates all three required values correctly.

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STEP 4: REQUIRED C PROGRAM

```
#include <stdio.h>
int main(void)
{
    int n;
    double side, apothem;
    double perimeter, area, interior_angle;

    printf("Enter number of sides, side length, and apothem: ");
    scanf("%d %lf %lf", &n, &side, &apothem);

    perimeter = n * side;
    area = 0.5 * perimeter * apothem;
    interior_angle = ((n - 2) * 180.0) / n;

    printf("Perimeter = %.2f units\n", perimeter);
    printf("Area = %.2f square units\n", area);
    printf("Each interior angle = %.2f degrees\n", interior_angle);

    return 0;
}
```

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STEP 5: COMPILATION, EXECUTION, AND EVALUATION:

```
┌──(kali㉿kali)-[~]
└─$ vi hdoc7.c

┌──(kali㉿kali)-[~]
└─$ cc hdoc7.c

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter number of sides, side length, and apothem: 5 4 2.75
perimeter = 20.00 units
Area = 27.50 square units
Each interior angle = 108.00 degrees

┌──(kali㉿kali)-[~]
└─$ 6 3 2.598
6: command not found

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter number of sides, side length, and apothem: 6 3 2.598
perimeter = 18.00 units
Area = 23.38 square units
Each interior angle = 120.00 degrees

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter number of sides, side length, and apothem: 8 5 6.036
perimeter = 40.00 units
Area = 120.72 square units
Each interior angle = 135.00 degrees

┌──(kali㉿kali)-[~]
└─$ 
```

Conclusion: Both C programs compile successfully, execute correctly, and produce the expected results for all prepared test cases.

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NAME - AYUSHI KUMAWAT

SAP ID – 590032252

BATCH – 16

COURSE – B.TECH CSE

THANKYOU